

The circadian modulation of melanin biosynthesis by targeting the ROR regulatory motif in the retinal pigment epithelium

Nemanja Milićević¹, Peppi Tolonen¹, Amandine Bery², Bobin George Abraham¹, Marie-Paule Felder-Schmittbuhl², Arthur A. Bergen^{3-5†}, Teemu O. Ihalainen¹, Soile Nymark¹

¹ Faculty of Medicine and Health Technology, Tampere University, Finland

² Centre National de la Recherche Scientifique, Université de Strasbourg, Institut des Neurosciences Cellulaires et Intégratives, 8 Allée du Général Rouvillois, F-67084 Strasbourg, France

³ Department of Human Genetics, Amsterdam University Medical Centers, location AMC, University of Amsterdam, Meibergdreef 9, 1105 AZ, Amsterdam, the Netherlands

⁴ Department of Ophthalmology, Amsterdam University Medical Centers, location AMC, University of Amsterdam, Meibergdreef 9, 1105 AZ, Amsterdam, the Netherlands

⁵ Emma Centre for Personalized Medicine, Amsterdam University Medical Centers, location AMC, University of Amsterdam, Meibergdreef 9, 1105 AZ, Amsterdam, the Netherlands

[†]The author passed away on September 15th 2024.

Introduction:

Pigmentation deficits are implicated in blinding eye disorders, with albinism being the most prominent. Melanin is the main pigment in the eyes, skin and/or hair and is synthesized in complex biochemical pathways. The circadian clock is a molecular pacemaker comprised of transcriptional and translation feedback loops involving transcription factors such as BMAL1, CLOCK, PER1-2, CRY1-2. Our goal was to investigate the link between the circadian clock and melanogenesis. We also utilized this link to modulate pigmentation in stem cell-derived pigmented cells of the eye, called the retinal pigment epithelium (RPE).

Methods:

We cultured embryonic stem cell-derived retinal pigment epithelium (hESC-RPE) for 12 weeks. The cells were synchronized by a serum-shock and harvested at 3h intervals for up to 66h. RT-PCR analysis was used to quantify the expression of clock and melanogenesis genes. We performed motif-enrichment search in melanogenesis genes by using online tools (*UCSC* database, Galaxy and TF Motif View). Pigmentation in hESC-RPE was studied by pharmacology, bright-field microscopy and analyzed by an AI-based trainable WEKA segmentation tool. Bioinformatics was used to analyze published RNA-sequencing data of laser capture microdissected mouse RPE obtained at 4 time-points over 24h.

Results:

We found that synchronized hESC-RPE show rhythmic mRNA expression of clock genes *BMAL1*, *CLOCK*, *CRY2*, *PER1*, *PER2*, *REV-ERB α* and melanogenesis-related genes *TYR*, *CDH3*, *DCT* and *PMEL*. We selected mouse and human conserved promoter regions by using the UCSC database with 6 known time-affected melanogenesis genes (*Tyr*, *Tyrp1*, *Pmel*, *Dct*, *Cdh3* and *Rab1a*). By TF Motif View, we found over-representation of binding sites for negative regulators of transcription and the ROR binding site, a known regulator of clock genes. Bright-field microscopy revealed decreased pigmentation in hESC-RPE treated with the ROR agonist Nobiletin compared to controls. Conversely, significantly increased pigmentation was observed in hESC-RPE cells supplemented with the indirect ROR antagonist SR9009 compared to controls. Finally, bioinformatics analysis revealed that melanogenesis pathways are enriched in genes up-regulated at night compared to late-afternoon time points in mouse RPE.

Conclusions:

Overall, these results suggest that melanogenesis is regulated by the circadian clock. Modulation of the clock could be a potential target for treating blinding pigment deficits such as albinism.

Keywords: retinal pigment epithelium, albinism, circadian clock, melanogenesis, stem cell